URIC ACID

Dr.maha



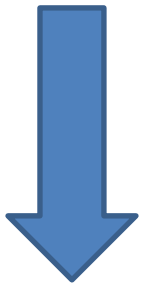
The GOUT!

Learning objectives

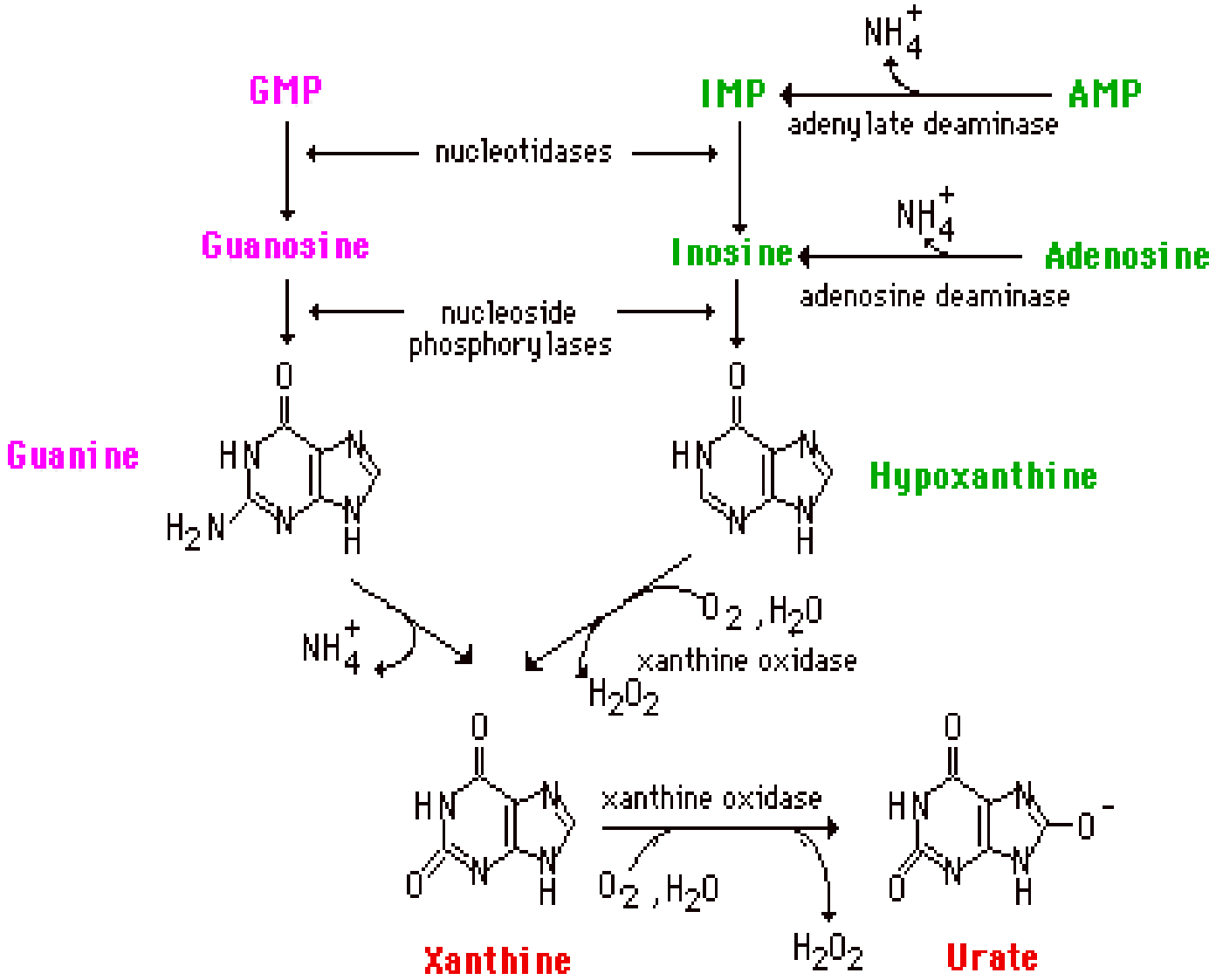
- Enlist the causes of hyperuricemia
- Describe clinical significance of hyper and hypouricemia
- Gout- causative factors
- Estimate uric acid level in the given blood sample.

Uric acid

- End product of Purine Catabolism
- Primarily excreted in the urine
- If accumulates in blood



HYPERURICEMIA


GMP
IMP
AMP
NH_4^+
adenylate deaminase
nucleotidases
Guanosine
Inosine
Adenosine
NH_4^+
adenosine deaminase
nucleoside
phosphorylases
Guanine
Hypoxanthine
NH_4^+
O_2 , H_2O
xanthine oxidase
H_2O_2
xanthine oxidase
O_2 , H_2O
H_2O_2
Xanthine
Urate

- Sources

❖Endogenous – daily synthesis rate from degradation of endogenous nucleic acid is 400mg

❖Dietary nucleic acid is converted to uric acid directly which contributes 300mg.

❖Total body pool

- Men- 1200mg
- Females- 600mg

Fate of uric acid in humans

- 75% is excreted in urine
- The remainder is secreted into the gastrointestinal tract to be excreted out in feces.

Clinical significance

?

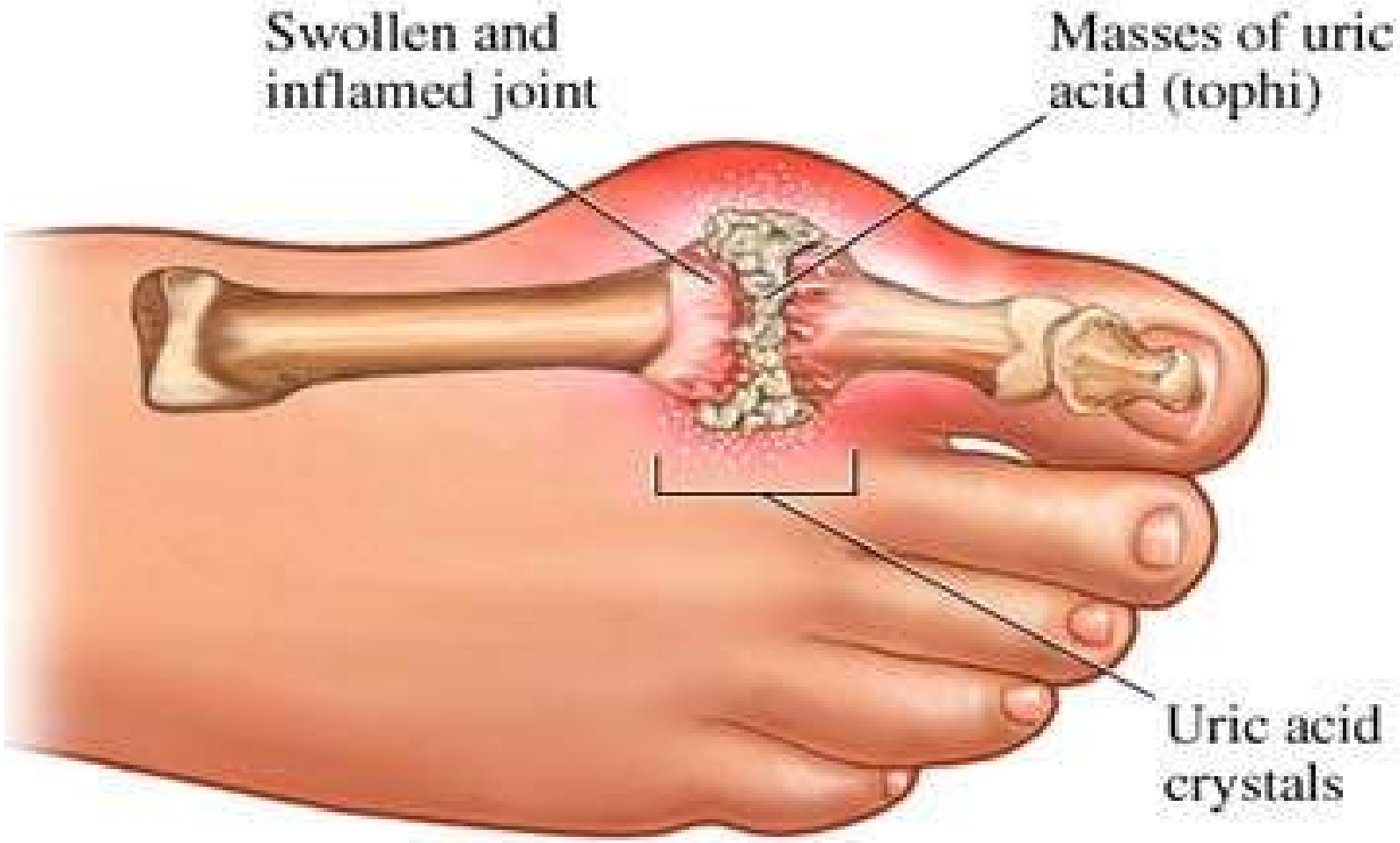
hyperuricemia

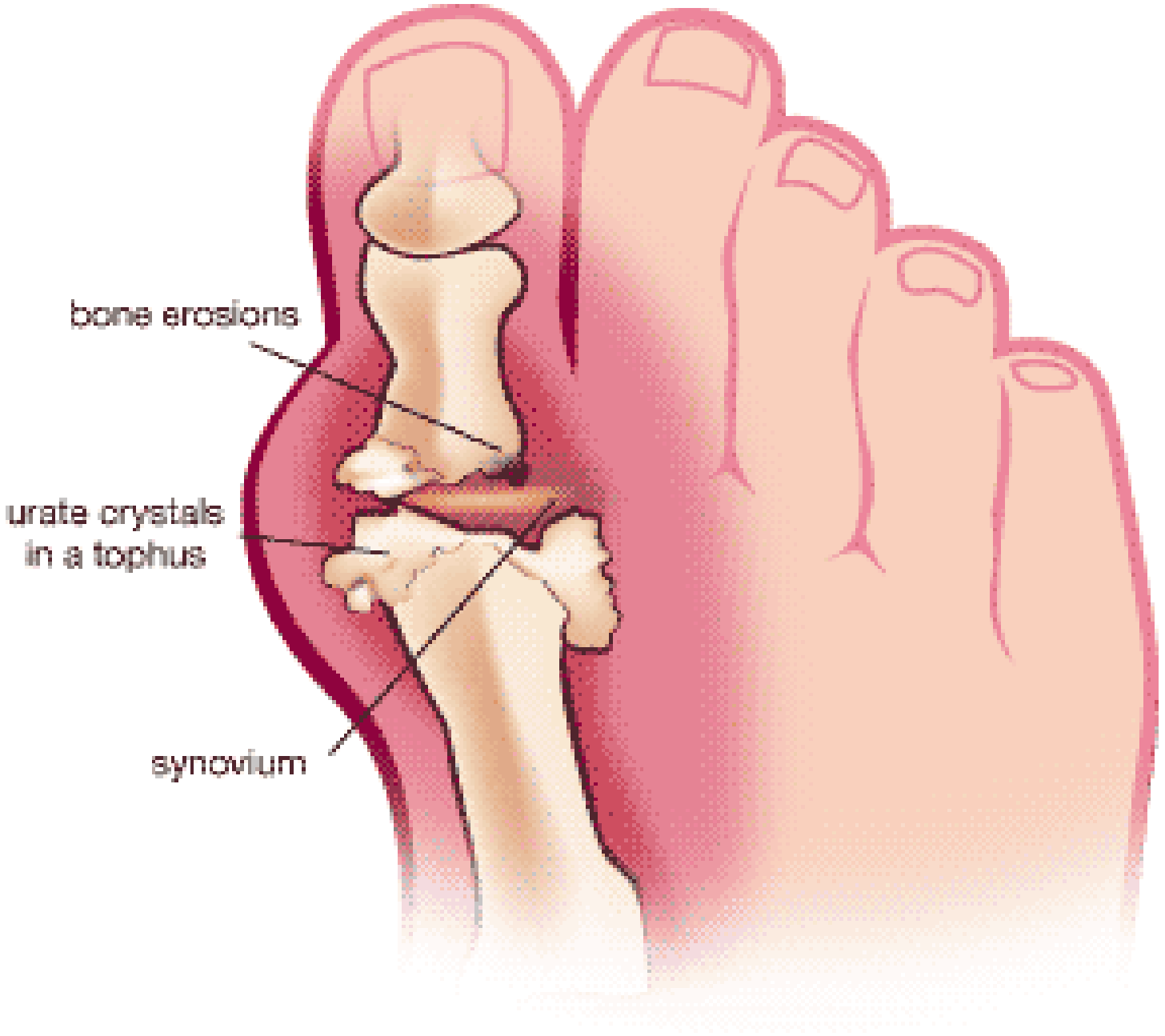
Primary causes	Secondary causes
Idiopathic	Excess dietary purine intake
Inherited metabolic disorders	Increased nucleic acid turnover
Psoriasis	Certain cancers (leukemia, lymphoma)
Altered ATP metabolism	certain forms of diabetes, or acidosis
Tissue hypoxia	
Pre- eclampsia	
Alcohol	

Gout

It is a group of disorders of nucleotide metabolism where crystals of **Monosodium urate monohydrate** derived from hyperuricemic body fluids give rise to inflammatory arthritis, tenosynovitis, tophaceous deposits and renal diseases.

What is gout


Swollen and inflamed joint
Masses of uric acid (tophi)
Uric acid crystals


bone erosions
urate crystals
in a tophus
synovium

Causes of Gout

- Under excretion of uric acid- common

- Overproduction of uric acid- less common



Purine nucleotide degradation

- Degradation of dietary nucleic acids occurs in the small intestine, where a
- family of pancreatic enzymes hydrolyzes the nucleic acids to nucleotides.
- Inside the intestinal mucosal cells, purine nucleotides are sequentially degraded by specific enzymes to nucleosides and free bases, with uric acid as the end product of this pathway. [Note: Purine nucleotides from denovo synthesis are degraded in the liver primarily. The free bases are sent out from liver and salvaged by peripheral tissues.]

Degradation in the small intestine

- Ribonucleases and deoxyribonucleases, secreted by the pancreas,
- hydrolyze dietary RNA and DNA primarily to oligonucleotides.
- Oligonucleotides are further hydrolyzed by pancreatic phosphodiesterases,
- producing a mixture of 3'- and 5'-mononucleotides. In the intestinal mucosal cells, a family of nucleotidases removes the phosphate groups hydrolytically, releasing nucleosides that are further degraded to free bases. Dietary purine bases are not used to any appreciable extent for the synthesis of tissue nucleic acids. Instead,they are generally converted to uric acid in intestinal mucosal cells.
- Most of the uric acid enters the blood, and is eventually excreted in
- the urine.

Uric acid formation

- [1] An amino group is removed from AMP to produce IMP by AMP deaminase, or from adenosine to produce inosine (hypoxanthineribose) by adenosine deaminase.
- [2] IMP and GMP are converted into their nucleoside forms—inosine and guanosine—by the action of 5'-nucleotidase.
- [3] Purine nucleoside phosphorylase converts inosine and guanosine

into their respective purine bases, hypoxanthine and guanine.

- [4] Guanine is deaminated to form xanthine.
- [5] Hypoxanthine is oxidized by xanthine oxidase to xanthine, which is further oxidized by xanthine oxidase to uric acid, the final product of human purine degradation. Uric acid is excreted primarily in urine

Diseases with purine degradation

- **Gout:** Gout is a disorder characterized by high levels of uric acid—
- the end product of purine catabolism—in blood (hyperuricemia),
- as a result of either the overproduction or underexcretion of uric acid. The hyperuricemia can lead to the deposition of mono - sodium urate crystals in the joints, and an inflammatory response to the crystals, causing first acute and then progressing to chronic gouty arthritis. Nodular masses of monosodium urate crystals (tophi) may be deposited in the soft tissues, resulting in chronic tophaceous gout . Formation of uric acid stones in the kidney (urolithiasis) may also be seen. [Note: Hyperuricemia is typically asymptomatic and does not lead to gout, but gout is preceded by hyperuricemia.] The definitive diagnosis of gout requires aspiration and examination of synovial fluid from an affected joint (or material from a tophus) using polarized light microscopy to confirm the presence of needle-shaped mono -sodium urate crystals